

Discrete Mathematics (MATH 208)

Fall 2022

Monday, 11/28/2022

Homework 13: Chapter 32

Score: / 10

Name (Print): _____

Chapter 32 - The Pigeonhole Principle

2 pts

1. Show that in a town with population 18,000, there must be at least two people with the same three initials.

2 pts

2. What is the smallest town population that will guarantee there will be at least two people with the same three initials?

2 pts

3. A developer buys a six mile by eight mile rectangular plot of land to build five houses. Show that no matter where the developer builds the houses, there will be at least two that are no more than five miles apart.

2 pts

4. Show that in any list of n integers, there will be a chunk of consecutive entries from the list that add up to a multiple of n . For example: in the list $-8, 4, 22, -11, 7$, we have $4 + 22 - 11 = 15$ is a multiple of 5.

2 pts

5. Suppose $a_1, a_2, a_3, \dots, a_{99}$ is a permutation of $1, 2, \dots, 99$. Show that the product

$$(a_1 + 1)(a_2 + 2)(a_3 + 3) \dots (a_{99} + 99)$$

is even.